Chapter: Chemical Reactions and Equations

1. Which of the statements about the reaction below are incorrect?
   $$2PbO(s) + C(s) \rightarrow 2Pb(s) + CO_2\ (g)$$
   a. Lead is getting reduced.
   b. Carbon dioxide is getting oxidised.
   c. Carbon is getting oxidised.
   d. Lead oxide is getting reduced.
      i. (a) and (b)
      ii. (a) and (c)
      iii. (a), (b) and (c)
      iv. all
2. $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
   The above reaction is an example of a
   a. combination reaction.
   b. double displacement reaction
   c. decomposition reaction
   d. displacement reaction
3. What happens when dilute hydrochloric acid is added to iron fillings? Tick the correct answer.
   a. Hydrogen gas and iron chloride are produced.
   b. Chlorine gas and iron hydroxide are produced
   c. No reaction takes place.
   d. Iron salt and water are produced.
4. What is a balanced chemical equation? Why should chemical equations be balanced?
5. Translate the following statements into chemical equations and then balance them.
   a. Hydrogen gas combines with nitrogen to form ammonia.
   b. Hydrogen sulphide gas burns in air to give water and sulpur dioxide.
   c. Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate.
   d. Potassium metal reacts with water to give potassium hydroxide and hydrogen gas.
6. Balance the following chemical equations.
   a. $HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$
   b. $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$
   c. $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
   d. $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$
7. Write the balanced chemical equations for the following reactions.
   a. Calcium hydroxide + Carbon dioxide → Calcium carbonate + Water
   b. Zinc + Silver nitrate → Zinc nitrate + Silver
   c. Aluminium + Copper chloride → Aluminium chloride + Copper
   d. Barium chloride + Potassium sulphate → Barium sulphate + Potassium chloride

8. Write the balanced chemical equation for the following and identify the type of reaction in each case.
    a. Potassium bromide(aq) + Barium iodide(aq) → Potassium iodide(aq) + Barium bromide(s)
    b. Zinc carbonate(s) → Zinc oxide(s) + Carbon dioxide(g)
    c. Hydrogen(g) + Chlorine(g) → Hydrogen chloride(g)
    d. Magnesium(s) + Hydrochloric acid(aq) → Magnesium chloride(aq) + Hydrogen(g)
9. What does one mean by exothermic and endothermic reactions? Give examples.
10. Why is respiration considered an exothermic reaction? Explain.
11. Why are decomposition reactions called the opposite of combination reactions? Write equations for these reactions.

Chapter: Acids, Bases and Salts

1. A solution turns red litmus blue, its pH is likely to be
    a. 1
    b. 4
    c. 5
    d. 10
2. A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains
    a. NaCl
    b. HCl
    c. LiCl
    d. KCl
3. 10 mL of a solution of NaOH is found to be completely neutralised by 8 mL of a given solution of HCl. If we take 20 mL of the same solution of NaOH, the amount HCl solution (the same solution as before) required to neutralise it will be
    a. 4 mL
    b. 8 mL
    c. 12 mL
    d. 16 mL
4. Which one of the following types of medicines is used for treating indigestion?
    a. Antibiotic
    b. Analgesic
    c. Antacid
    d. Antiseptic
5. Write word equations and then balanced equations for the reaction taking place when
    a. dilute sulphuric acid reacts with zinc granules.
    b. dilute hydrochloric acid reacts with magnesium ribbon.
    c. dilute sulphuric acid reacts with aluminium powder.
    d. dilute hydrochloric acid reacts with iron filings.
6. Compounds such as alcohols and glucose also contain hydrogen but are not categorised as acids. Describe an Activity to prove it.
7. Why does distilled water not conduct electricity, whereas rain water does?
8. Why does dry HCL gas not show acidic behaviour in the absence of water?

9. Five solutions A,B,C,D and E when tested with universal indicator showed pH as 4,1,11,7 and 9, respectively. Which solution is
    a. neutral?
    b. strongly alkaline?
    c. strongly acidic?
    d. weakly acidic?
    e. weakly alkaline?
    Arrange the pH in increasing order of hydrogen-ion concentration.
10. Equal lengths of magnesium ribbons are taken in test tubes A and B. Hydrochloric acid (HCl) is added to test tube A, while acetic acid (CH_3COOH) is added to test tube B. Amount and concentration taken for both the acids are same. In which test tube will the fizzing occur more vigorously and why?
11. Fresh milk has a pH of 6. How do you think the pH will change as it turns into curd? Explain your answer.
12. A milkman adds a very small amount of baking soda to fresh milk. (a) Why does he shift the pH of the fresh milk from 6 to slightly alkaline? (b) Why does this milk take a long time to set as curd?
13. Plaster of Paris should be stored in a moisture-proof container. Explain why?
14. What is a neutralisation reaction? Give two examples.
15. Give two important uses of washing soda and baking soda.

Chapter: Metals and Non-metals

1. Which of the following pairs will give displacement reactions?
    a. NaCl solution and copper metal
    b. $MgCl_2$ solution and aluminium metal
    c. $FeSO_4$ solution and silver metal
    d. $AgNO_3$ solution and copper metal.
2. Which of the following methods is suitable for preventing an iron frying pan from rusting?
    a. Applying grease
    b. Applying paint
    c. Applying a coating of zinc
    d. All of the above.
3. An element reacts with oxygen to give a compound with a high melting point. This compound is also soluble in water. The element is likely to be
    a. calcium
    b. carbon
    c. silicon
    d. iron.
4. Food cans are coated with tin and not with zinc because
    a. zinc is costlier than tin.
    b. zinc has a higher melting point than tin.
    c. zinc is more reactive than tin.
    d. zinc is less reactive than tin.

5. You are given a hammer, a battery, a bulb, wires and a switch.
    a. How could you use them to distinguish between samples of metals and non-metals?
    b. Assess the usefulness of these tests in distinguishing between metals and non-metals.
6. What are amphoteric oxides? Give two examples of amphoteric oxides.
7. Name two metals which will displace hydrogen from dilute acids, and two metals which will not.
8. In the electrolytic refining of a metal M, what would you take as the anode, the cathode and the electrolyte?
9. State two ways to prevent the rusting of iron.
10. What type of oxides are formed when non-metals combine with oxygen?
11. Give reasons
    a. Platinum, gold and silver are used to make jewellery.
    b. Sodium, potassium and lithium are stored under oil.
    c. Aluminium is a highly reactive metal, yet it is used to make utensils for cooking.
    d. Carbonate and sulphide ores are usually converted into oxides during the process of extraction.
12. You must have seen tarnished copper vessels being cleaned with lemon or tamarind juice. Explain why these sour substances are effective in cleaning the vessels.
13. Differentiate between metal and non-metal on the basis of their chemical properties.
14. A man went door to door posing as a goldsmith. He promised to bring back the glitter of old and dull gold ornaments. An unsuspecting lady gave a set of gold bangles to him which he dipped in a particular solution. The bangles sparkled like new but their weight was reduced drastically. The lady was upset but after a futile argument the man beat a hasty retreat. Can you play the detective to find out the nature of the solution he had used?
15. Give reasons why copper is used to make hot water tanks and not steel (an alloy of iron).